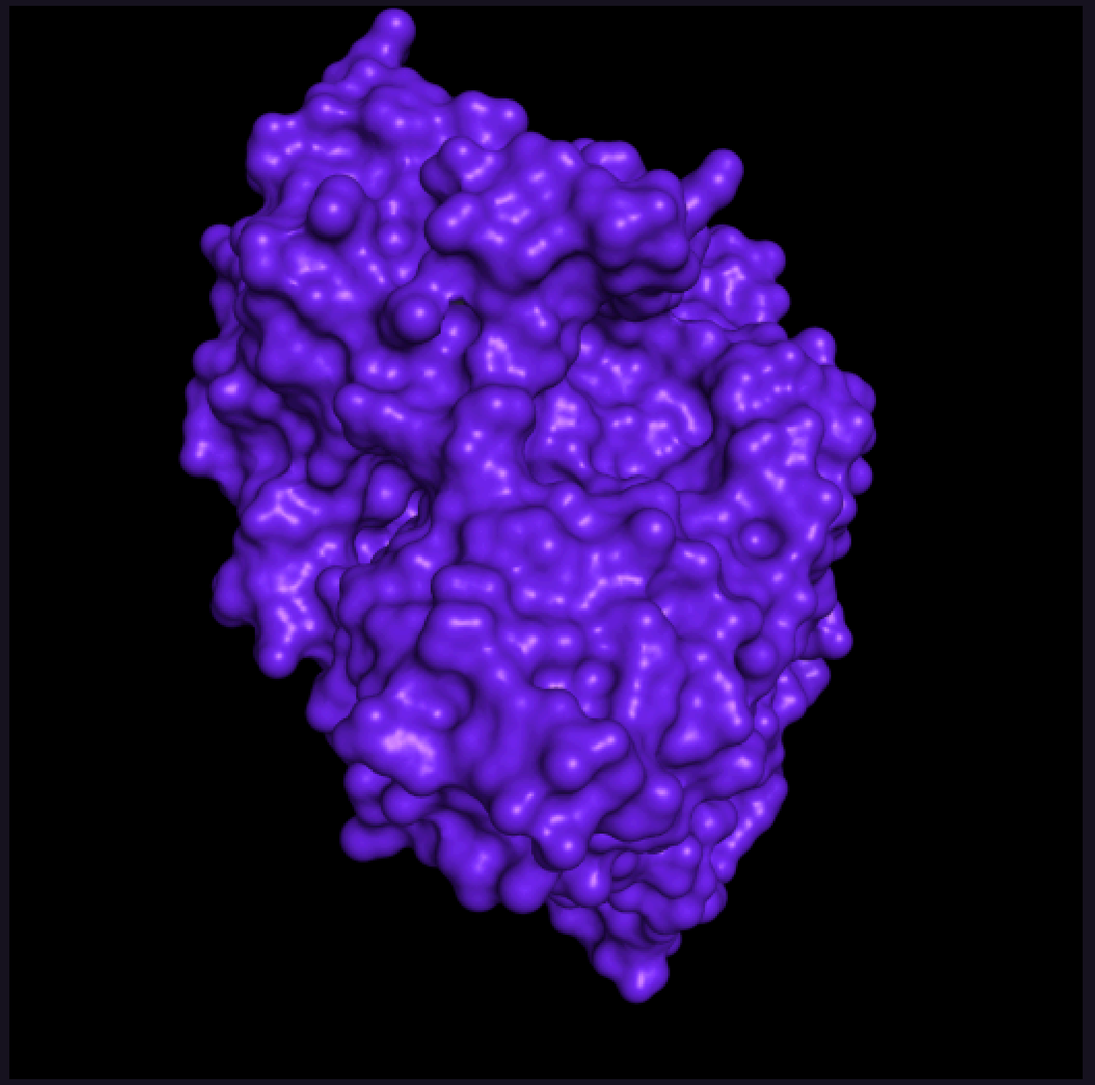
BIO 4525 WORKSHOP

PyMOL scenes and movies

Tell a protein’s story in PyMOL: scenes, camera moves, fades and moving molecules.

github.com/anshuJS/Tutorials_BIOL4525



WHERE THIS STARTED

A 4 minute movie about one enzyme



Watch it on YouTube

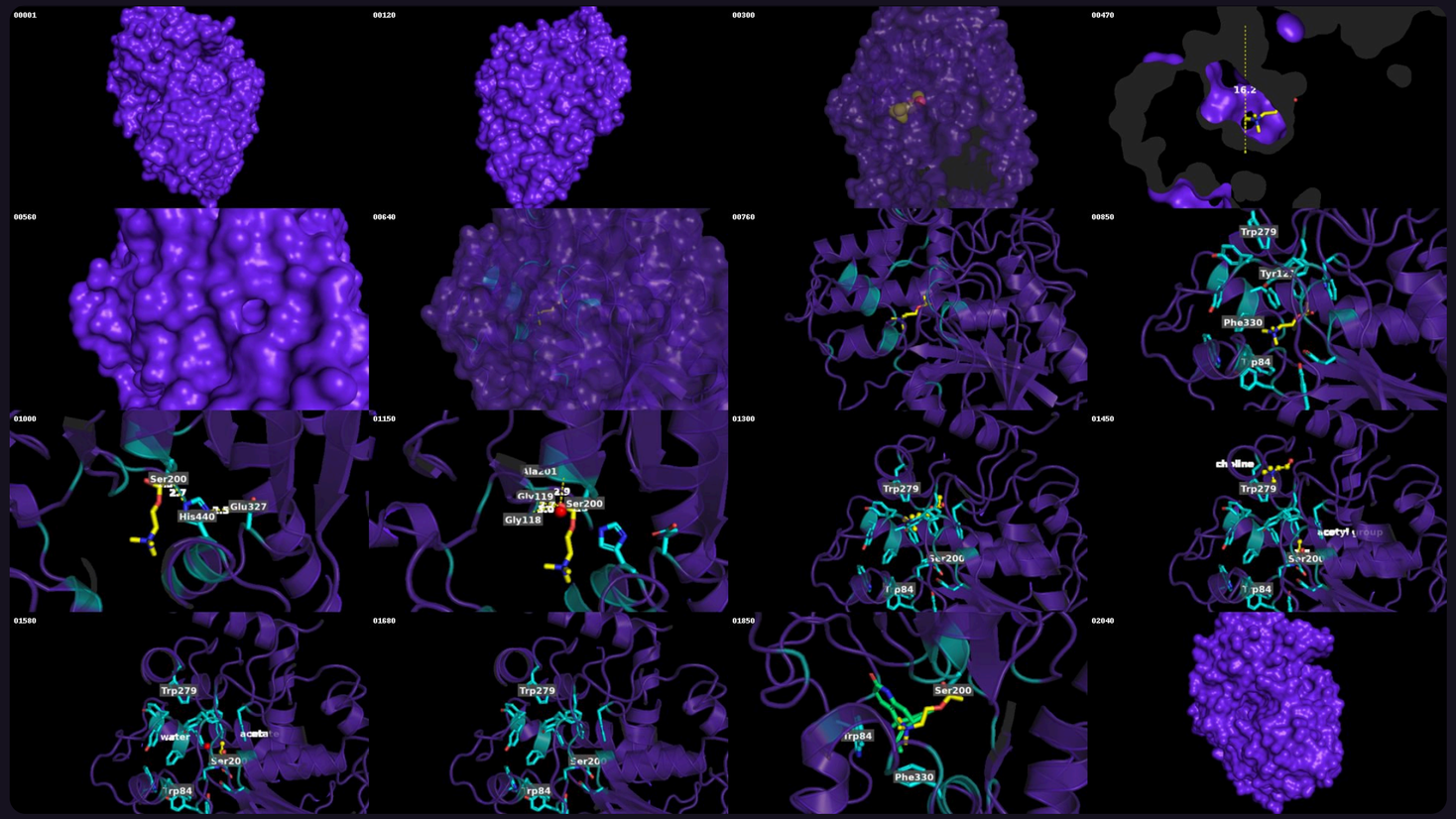
youtu.be/sA3F8CdAZTY

A class project on acetylcholinesterase: PyMOL scenes, plus Blender and video editing.

Today we rebuild its story **using only PyMOL**.

TODAY

The same story, made only in PyMOL



68 seconds, acetylcholinesterase. No other software.

Every shot uses one PyMOL feature you will learn today.

Then you do the same with **your own paper figure**.

THE PLAN

90 minutes, one skill at a time

Minutes	Part	You will
0 to 8	Watch	see the finished movie
8 to 15	Step 1	load and style the protein
15 to 30	Step 2	make scenes
30 to 47	Steps 3 and 4	your first movies: roll, rock, keyframes
47 to 55	Steps 5 and 6	watch the demo: fades and moving molecules
55 to 80	Step 9	turn YOUR paper figure into a movie, in teams
80 to 90	Show and tell	play your movie, save it

SETUP • 3 MIN

Get the files

IN A BROWSER

1. Go to github.com/anshuJS/Tutorials_BIOL4525
2. Green **Code** button, then **Download ZIP**
3. Unzip it on your Desktop

The workshop is the folder **PyMOLMovies** inside **Tutorials_BIOL4525-main**.

THEN TYPE IN PYMOL

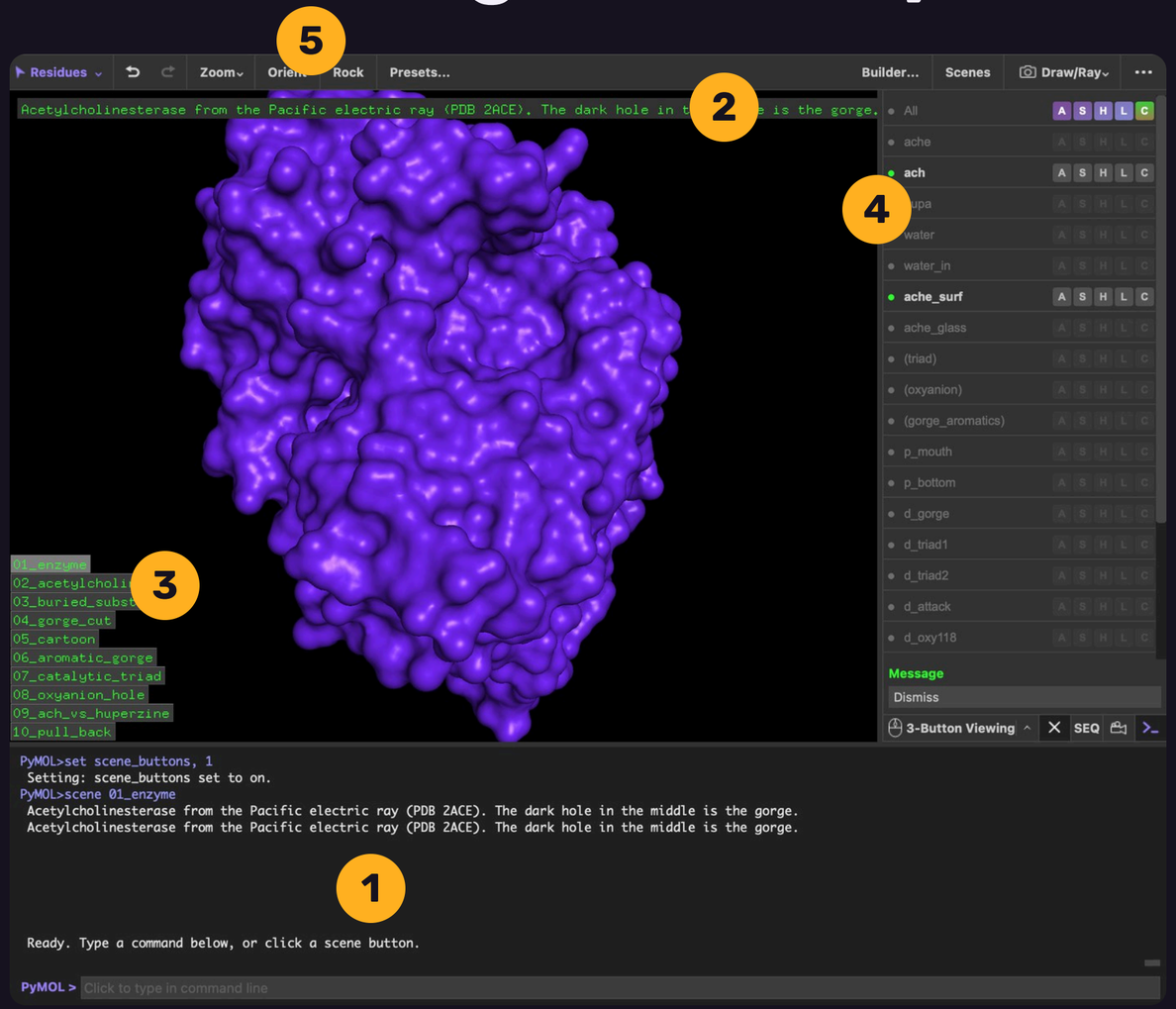
```
cd ~/Desktop/Tutorials_BIOL4525-
main/PyMOLMovies

ls
```

> **Check:** ls lists steps, structures and checkpoints.

Windows: cd C:/Users/YOURNAME/Desktop/Tutorials_BIOL4525-main/PyMOLMovies • Already use git? git clone works too.

Where things are in PyMOL 3.1



1 Command line, at the bottom. Type everything here.

2 Scene caption, across the top of the viewer.

3 Scene buttons. Click one to jump to it.

4 Objects. A green dot means on. Click a name to switch it.

5 Toolbar: Rock, Scenes, Draw/Ray.

Movie controls appear in a **Global Frames** panel at the bottom right once you make a movie.

READING THE CODE

A .pml file is just commands, one per line

```
# a comment. PyMOL skips it
```

Explains the next lines. Never put a ; in one

```
load structures/2ace.pdb, raw2ace
```

Command, then its arguments, separated by commas

```
color cyan, ache and resi 200
```

The last argument is often a **selection**: which atoms

```
show sticks, triad and not name N+C+O
```

Combine with and / or / not. N+C+O means any of them

```
@steps/02_scenes.pml
```

@ runs a whole file, line by line

Selection words you will see: resi (residue number), resn (residue name), name (atom name), chain, within.

STEP 1 • 7 MIN

Load the protein and its actors

TYPE

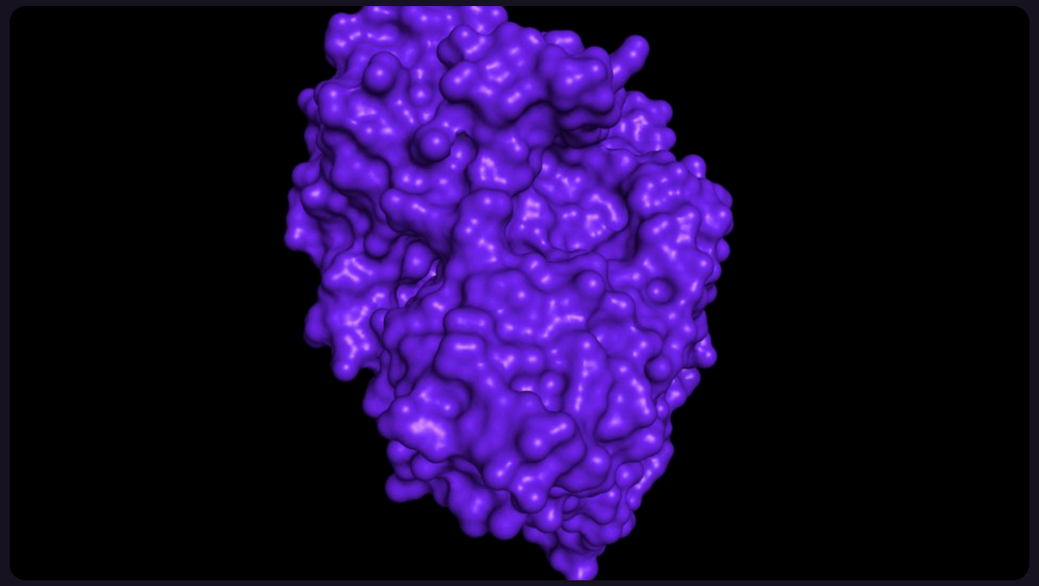
```
@steps/01_load_and_style.pml
```

THEN TRY

Click the object names on the right: **ache**, **ach**, **hupa**, **ache_glass**. Each one is an actor you can switch on and off.

> **Check:** a purple surface, and a long list of objects.

YOU SHOULD SEE



The idea: one object per actor. Separate objects give you control in every later step.

INSIDE STEP 1

Build the actors, then dress them

```
create ache, raw2ace and polymer
```

Copy the protein into its own object

```
create ach, raw2ace and resn ACH
```

The substrate gets its own object

```
set_color ache_purple, [0.45, 0.15, 0.95]
```

Invent a colour: red, green, blue from 0 to 1

```
util.cnc ache
```

Colour N blue and O red. Carbons keep their colour

```
set transparency, 0.6, ache_glass
```

A setting on ONE object (scenes will not remember it)

```
distance d_triad1, ...resi 200 and name OG, ...resi 440 and
name NE2
```

A dashed line with the distance in A

```
pseudoatom lab_triad, ache and resi 200, label=Ser200
```

An invisible point carrying a label

STEP 2 • SCENES

A scene is a snapshot you can jump back to

It remembers

- the camera view
- which objects are on
- cartoon, sticks, surface…
- colours
- a caption (message=)

It forgets

- label text
- settings: transparency, sphere size, background

So step 1 made separate label objects and a separate see-through surface.

STEP 2 · BY HAND · 7 MIN

Make two scenes yourself

TYPE, ONE LINE AT A TIME

```
orient ache_surf
scene my_first, store, message=The whole enzyme
disable ache_surf
enable ach
show sticks, ach
zoom ach, 4
scene my_second, store, message=The substrate
```

THEN

Press **Page Up** and **Page Down**, or click the two new scene buttons.

The camera flies, objects switch, the caption changes.

> **Check:** two buttons, my_first and my_second.

Captions: no commas inside message=. The caption stops at the first comma.

STEP 2 • THE STORY • 5 MIN

Load all ten scenes of the story

TYPE

```
@steps/02_scenes.pml
```

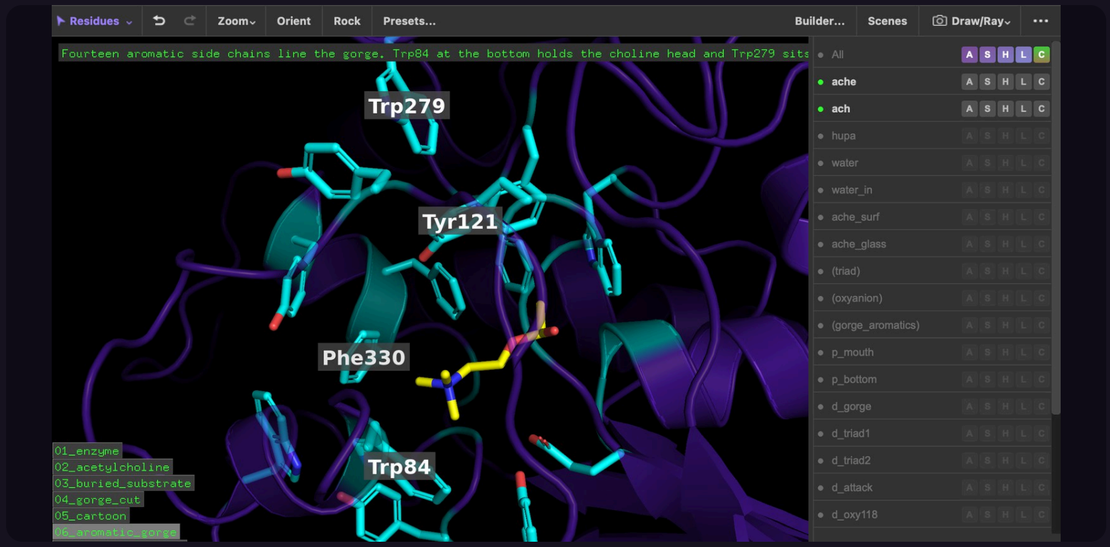
THEN TRY

Click through the scene buttons in order. Read each caption.

```
set scene_animation_duration, 3
```

Check: buttons 01_enzyme to 10_pull_back.

YOU SHOULD SEE



Behind? load checkpoints/after_step2.pse and you are caught up.

INSIDE STEP 2

Every scene is the same four moves

```
# 1 switch off what you do not need
disable all
hide everything, ache or ach or hupa
# 2 switch on and show what you need
enable ache
show cartoon, ache
show sticks, triad and not name N+C+O
enable lab_triad
# 3 place the camera
set_view (... 18 numbers ...)
# 4 store it, with a caption
scene 07_catalytic_triad, store, message=...
```

The 18 numbers in set_view

1 to 9 which way the camera looks

10 to 12 how far away it is (zoom)

13 to 15 the point it orbits around

16 and 17 front and back clipping planes

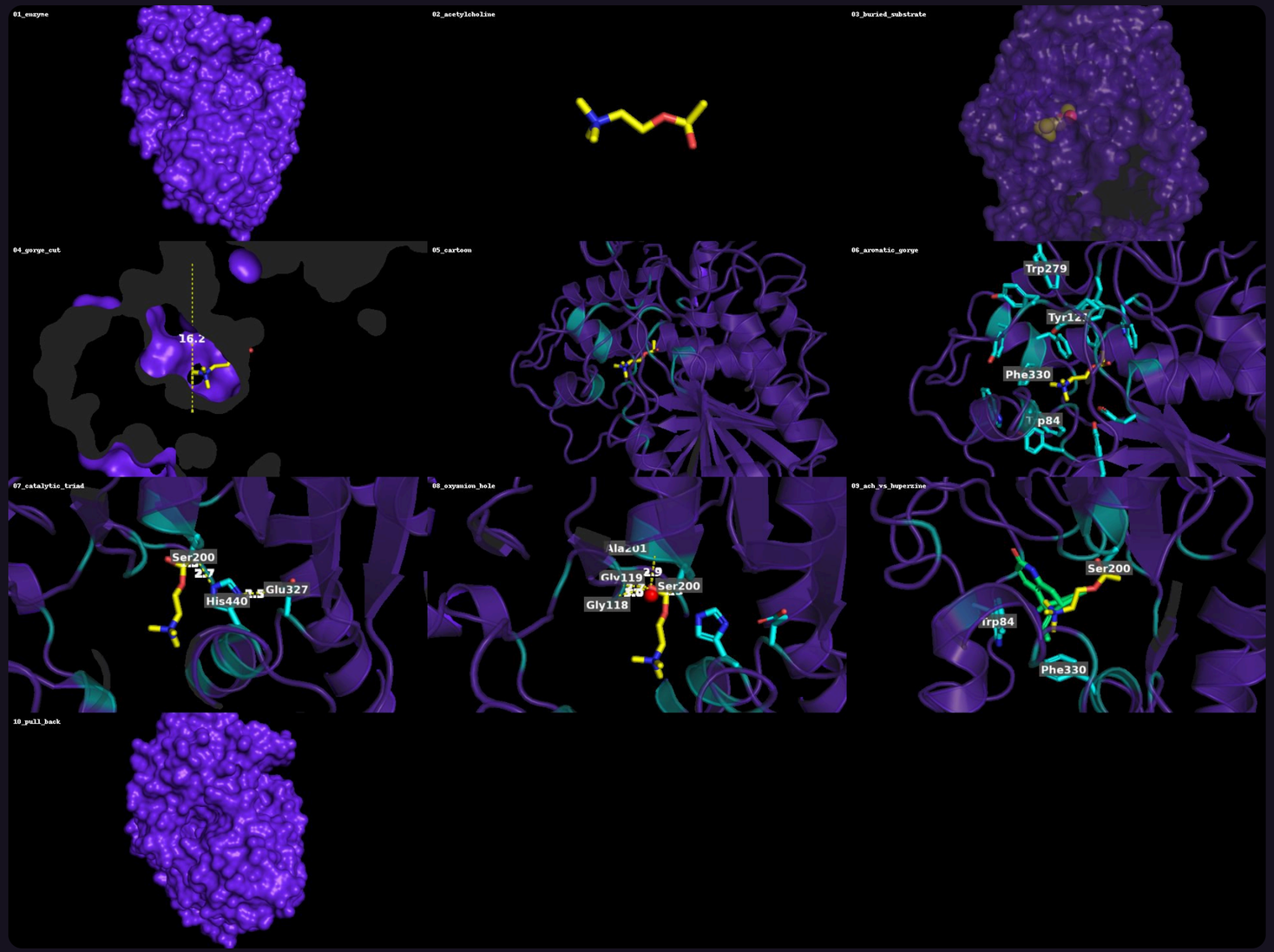
18 perspective or flat

Never type them. Move with the mouse, type **get_view**, copy what it prints.

THE STORY IN TEN SCENES

Wide, closer, the answer, back out

Enzyme, substrate, where it hides, the gorge, the aromatic lining, the catalytic triad, the oxyanion hole, a real drug, pull back.



STEP 3 · 7 MIN

Your first movie: roll and rock

TYPE

```
@steps/03_roll_and_rock.pml
```

Press **play** in the Global Frames panel (bottom right), or type mplay. Stop with mstop.

THEN A ROCK ON THE TRIAD

```
scene 07_catalytic_triad, animate=0
mset
movie.add_rock(6, angle=30, axis="y", start=1)
```

KNOW THIS

A movie is a list of frames. 30 frames = 1 second.

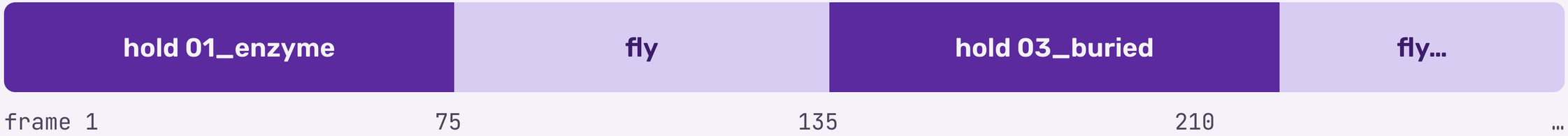
A camera movie films whatever is on screen. Recall a scene first, then build the movie.

> **Check:** the roll is 240 frames, 8 seconds.

Same as the menu: Movie, Program, Camera Loop. The 2023 AChE video opened with exactly this roll.

STEP 4 • KEYFRAMES

Tell PyMOL a few frames. It fills in the rest.




```
mview store, 1, scene=01_enzyme
mview store, 75, scene=01_enzyme
mview store, 135, scene=03_buried_substrate
mview store, 210, scene=03_buried_substrate
```

Same scene twice = **hold**.

Different scenes = **fly**.

Hold about 2.5 s (75 frames) so people can read.

STEP 4 • 10 MIN

A 43 second movie made of scenes

TYPE, THEN PLAY

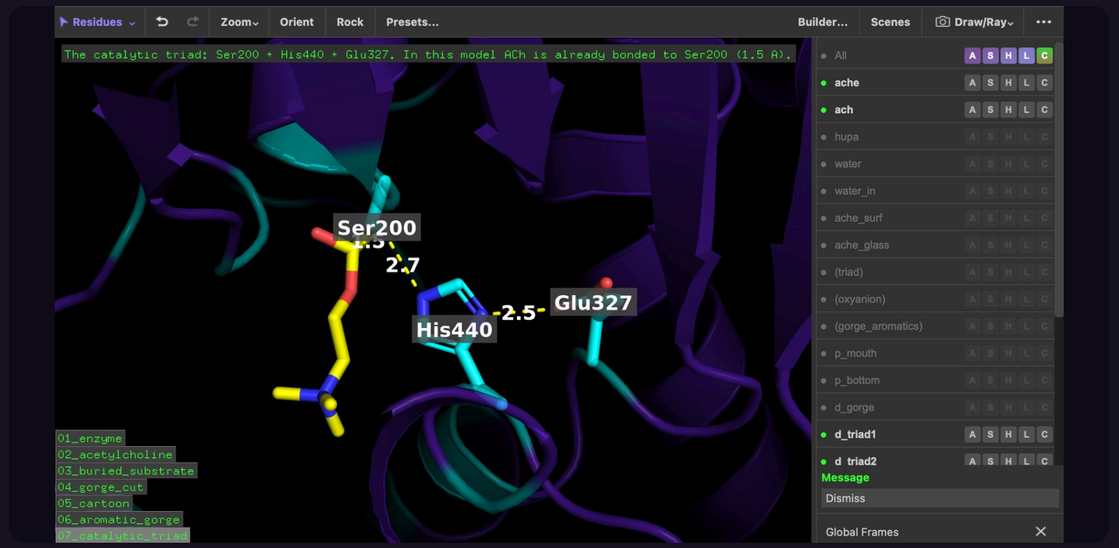
```
@steps/04_scene_movie.pml
```

THEN CHANGE IT

1. Make the triad shot longer
2. Skip a scene: mview clear, 270
3. Add 02_acetylcholine somewhere

> **Check:** the frame counter says 1290.

YOU SHOULD SEE (FRAME 700)



Camera stuck? A movie with keyframes owns the camera. Type mset to delete the movie.

INSIDE STEPS 3 AND 4

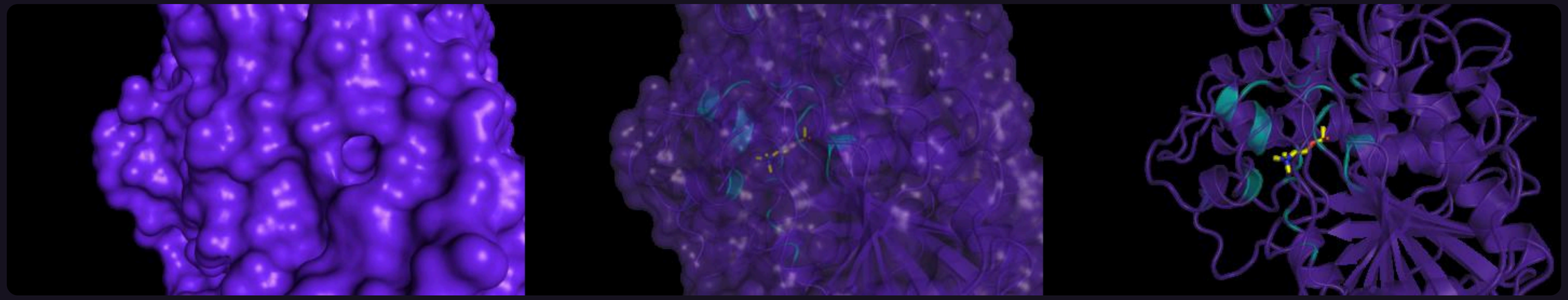
Frames, then keyframes

`set movie_fps, 30`	30 frames make 1 second
`mset 1 x1290`	Make 1290 empty frames: 43 s
`mview store, 135, scene=03_buried_substrate`	At frame 135 be in this scene. PyMOL flies the frames in between
`movie.add_roll(8, axis="y", start=1)`	Shortcut: makes the frames and a full turn in one go
`frame 1 / mplay / mstop`	Go to a frame, play, stop
`mset`	Nothing after it: delete the movie

Seconds × 30 = frames. A 2.5 s hold is 75 frames. A 2 s flight is 60 frames.

STEP 5 • WATCH THE DEMO

Fade the surface away, frame by frame



```
mdo 60: set transparency, 0.3, ache_fade
mdo 61: set transparency, 0.31, ache_fade
… one per frame, written by a short loop
```

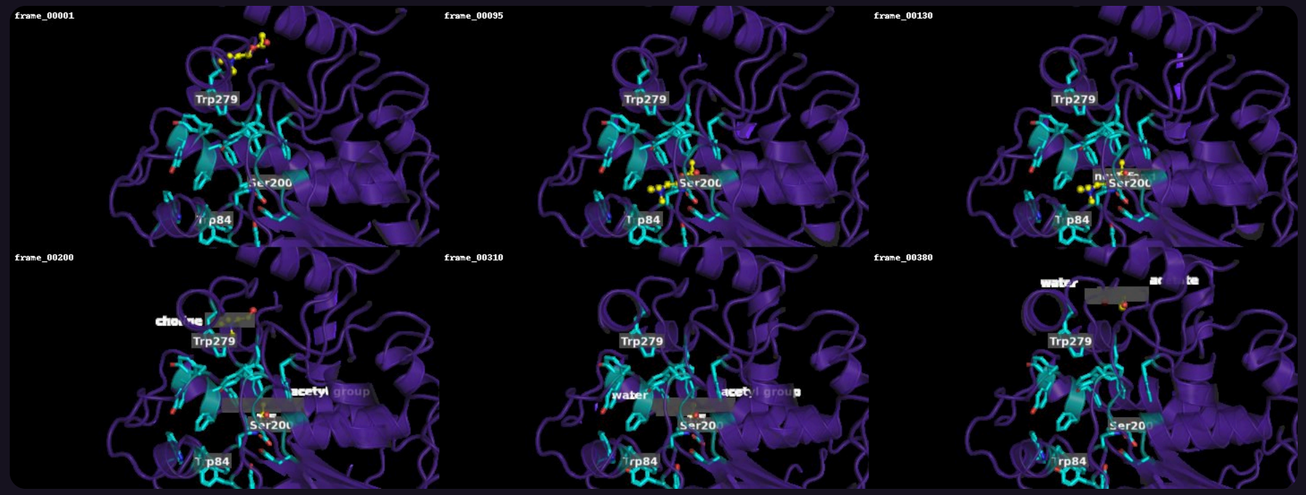
Scenes cannot fade: transparency is a setting.

mdo runs a command when the movie reaches a frame.

Try it later: @steps/05_surface_fade.pml • In the 2023 video this shot was made in Blender.

STEP 6 • WATCH THE DEMO

Molecules can have keyframes too



ACh slides down the gorge. Ser200 bonds it. Choline leaves. Water comes in. Acetate leaves.

A cartoon of the chemistry, not a simulation.

```
set movie_auto_store, 0
translate [-6.0, -15.3, 7.3], object=m_ach, camera=0
mview store, 1, object=m_ach start at the gorge mouth
matrix_reset m_ach, mode=1
mview store, 90, object=m_ach arrive in the active site
```

Try it later: @steps/06_mechanism.pml • Also a Blender shot in the 2023 video, now pure PyMOL.

INSIDE STEPS 5 AND 6

Commands on frames, and objects that move

The fade (step 5)

```
python
for frame_no in range(1, 181):
t = (frame_no - 30) / 90.0
t = min(max(t, 0.0), 1.0)
cmd.mdo(frame_no, "set transparency, %.3f, ache_fade" %
t)
python end
```

For each of the 180 frames, work out how see-through the surface should be: 0 until frame 30, rising to 1 by frame 120. **mdo** pins that command to the frame.

The moving ACh (step 6)

```
set movie_auto_store, 0
translate [-6.0, -15.3, 7.3], object=m_ach, camera=0
mview store, 1, object=m_ach
matrix_reset m_ach, mode=1
mview store, 90, object=m_ach
mdo 150: disable m_ach
```

Push ACh 18 A up the gorge and remember that at frame 1. Put it back home and remember that at frame 90. PyMOL slides it between. At frame 150, swap it for its two halves.

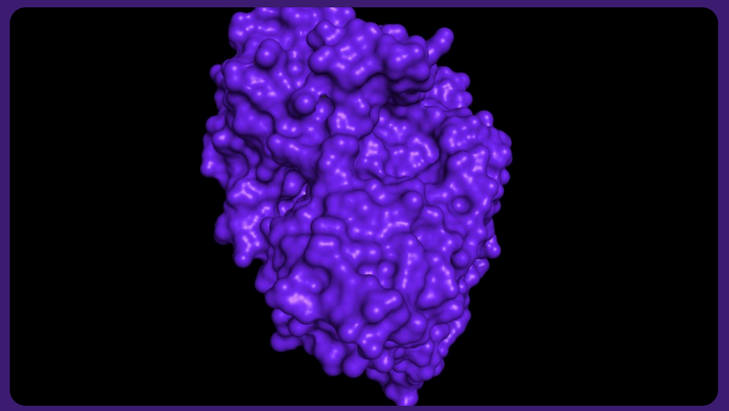
BEFORE YOU START ON YOUR OWN

Things that trip people up

If this happens	Do this
A label or transparency changed in every scene	Scenes forget those: use separate objects
Caption cut short	No commas inside message=
Strange error from a comment line	No ; anywhere in a .pml, even after #
The camera will not stay where you put it	The movie owns it: type mset
A moving object ignores your keyframe	set movie_auto_store, 0 first
Saved movie will not play after loading	Type accept at the security warning

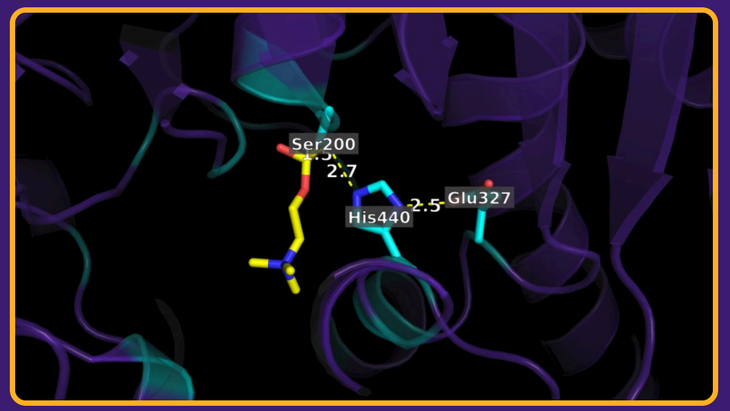
STEP 9 • YOUR TURN • 25 MIN, IN PROJECT TEAMS

Your paper figure script is already a scene



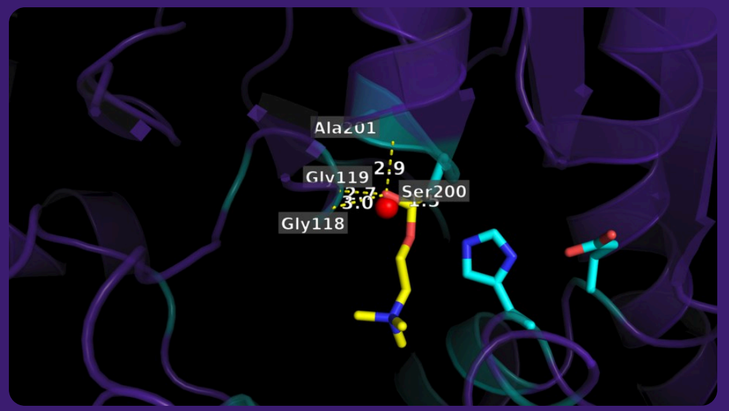
1 Wide

the whole protein



2 Your figure

exactly as in your paper



3 Close-up

what the figure is about, then a gentle rock

12 seconds. Examples shown with AChE. You use your own protein and your own figure script.

STEP 9 • PREPARE • 5 MIN

Get your figure script ready

1. Find the .pml your team uploaded with your figures
2. Copy it. Open the copy in a text editor
3. Put a # in front of every **ray** and **png** line
4. A **load** line points to someone else's laptop? Fix the path, or use **fetch** with the PDB code

BEFORE AND AFTER

```
ray
png figure1.png
becomes
# ray
# png figure1.png
```

Check: run it alone and your figure appears.

Test it on its own first: @path/to/your_figure.pml

STEP 9 • MAKE IT • 20 MIN

Fill in the blanks in step 9

Open **steps/09_your_figure_movie.pml** in a text editor. Fill every ____

```
@____/your_figure.pml
your edited script
message=____
three captions, no commas
zoom ____, 3
small, chain named: chain A and resn NAP
```

THEN

```
@steps/09_your_figure_movie.pml
```

Press play. Change the keyframe numbers until the timing feels right. Run again.

> **Check:** three scene buttons and 390 frames.

A select line in your own script often gives you the close-up for free: zoom pocket, 3

STEP 9 • STRETCH • DONE EARLY?

Two panels, two scenes, one flight

IN A COPY OF PANEL B'S SCRIPT, PUT # IN FRONT OF

- delete all / reinitialize
- fetch or load of something already loaded
- remove lines

Panel B must not undo panel A. Colours are safe: each scene keeps its own.

```
@path/panelA.pml
scene panel_A, store
@path/panelB_edited.pml
scene panel_B, store
mset 1 x240
mview store, 1, scene=panel_A
mview store, 75, scene=panel_A
mview store, 165, scene=panel_B
mview store, 240, scene=panel_B
```

Background colour is a setting: the last script to set it wins for both panels.

SHOW AND TELL • THEN STEP 7

Save first. Render at home.

1 Now

```
save my_figure_movie.pse
```

Keeps every scene and keyframe. You can change the movie next week.

2 Licensed PyMOL

File, Export Movie As, MPEG

Writes .mpg. VLC plays it. MP4 needs ffmpeg.

3 Any PyMOL

```
mpng frames/frame_
```

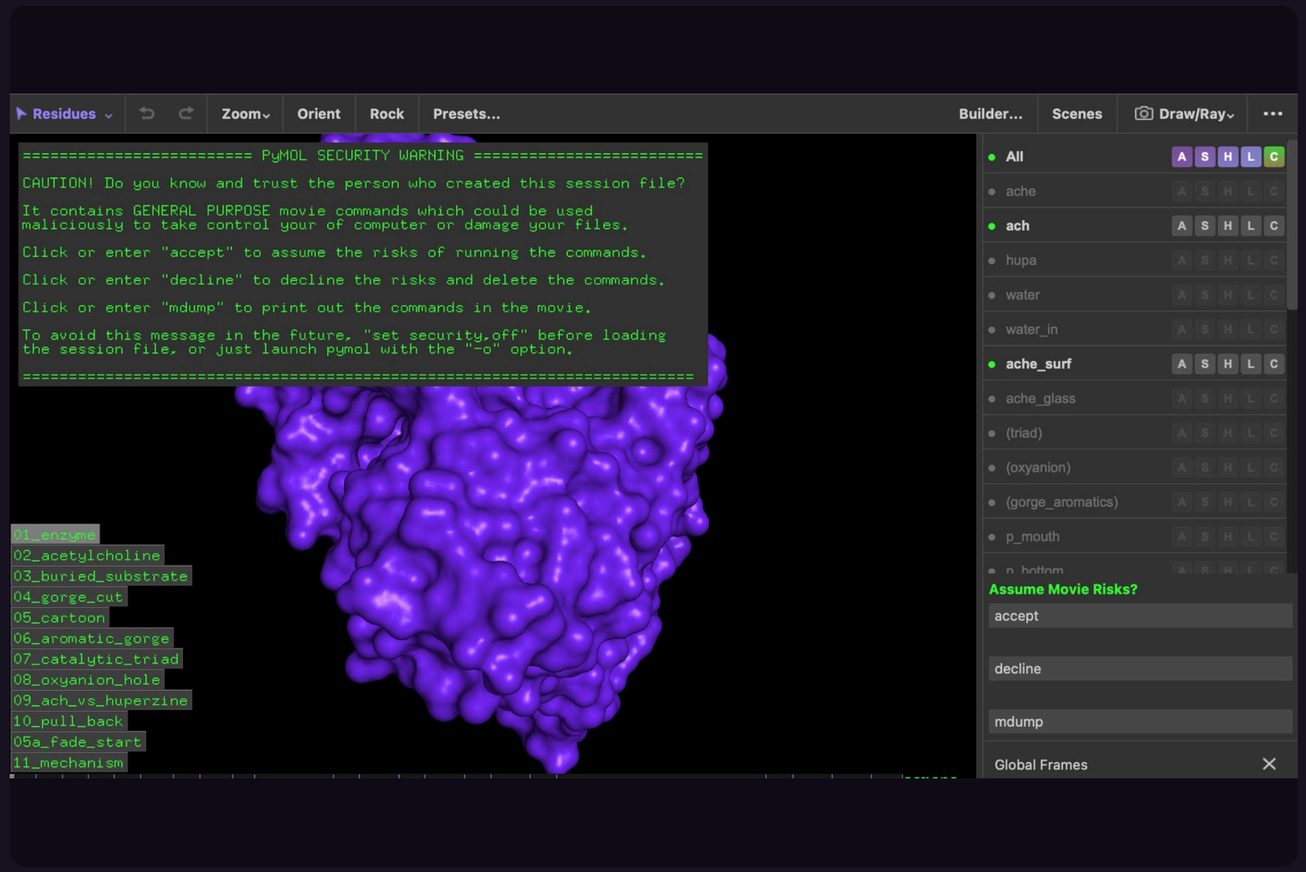
One PNG per frame. Join them in Fiji or any image-sequence tool.

Each team plays its movie from the front: **30 seconds each**.

A ray-traced 40 s movie can take an hour. Test small first: mode="draw", 640 x 360.

OPENING A SAVED MOVIE

A security warning is normal. Type accept.



Sessions with **mdo** commands (fades, moving objects) ask first.

Until you accept, the movie will not play.

```
accept
```

Try it: load sessions/AChE_workshop_final.pse, accept, mplay.

Everything is in the repo

`github.com/anshuJS/Tutorials_BIOL4525 › PyMOLMovies`

Handout

every command and the traps, on one page

Optional step 8

a 15 s movie: how does huperzine A block the enzyme?

Your figure movie

a great extra for your lightning talk